An Automated Urban Growth Analysis and Growth Direction Prediction Pipeline Using Multi-Temporal Landsat Imagery

Anshu Budha chhetri
[1]*Department of Computer Science, Georgia State University*
Atlanta, Georgia, United States
Abudhachhetri1@student.gsu.edu

***Abstract*— Accurate detection of urban land cover change over multi-decadal periods requires consistent, globally applicable classification techniques and methods that do not rely on hand-labelled training data or manual threshold tuning for each interested area independently. This paper presents an automated, four-stage pipeline that combines two peer-reviewed global products — the JRC Global Human Settlement Layer (GHSL) for epochs prior to 2015, and Google Dynamic World for epochs from 2015 onwards — to produce land cover classification and change maps for any user-defined area of interest (AOI). The pipeline ingests data directly from Google Earth Engine, performs automatic coordinate reference system selection, remaps class vocabularies to a common seven-class scheme, and outputs georeferenced rasters, urban expansion maps, directional growth roses, kernel density heatmaps, and change statistics. Validation on three different Districts, Nepal (1985–2024) – (Rupandehi, Banke, Bardiya) yields a measured urban growth (331% Rupandehi, 251.8% Banke and 74.2% Bardiya) corresponding to a compound annual growth rate consistent with independent estimates from GHSL 2020 and published Terai urbanisation literature. The pipeline is configurable for any global AOI by editing four parameters.**

Key words — urban expansion, land cover change detection, GHSL, Dynamic World, Google Earth Engine, remote sensing, change mapping.

I. INTRODUCTION

Urban areas are among the fastest-changing landscape types on Earth, with global built-up surfaces expanding rapidly in many regions [1]. Monitoring these changes is essential for infrastructure planning, environmental management, flood risk assessment, and land-use policy evaluation. Remote sensing remains the most practical approach for analysing such changes across large geographic areas and long time periods.

However, performing multi-temporal land cover analysis remains challenging, especially for users with requirements of surficial information. Traditional approaches often require collecting labelled training data, selecting appropriate classification methods, and ensuring consistency across different time periods. These steps can be time-consuming, technically complex, and difficult to reproduce across different study areas.

To address this gap, this paper presents a simplified and accessible pipeline designed for users from diverse academic backgrounds. The pipeline enables users to perform built-up area change analysis with minimal technical knowledge by simply providing a region of interest. It automates data selection, processing, alignment and comparison, producing consistent outputs including classified maps and change analysis results.

The proposed approach leverages globally available, peer-reviewed datasets suited for different time periods, ensuring consistency without requiring manual training or parameter tuning. By minimising the technical overhead typically associated with geospatial workflows and complex toolchains, the pipeline makes remote sensing–based change detection more accessible and reproducible for a broader audience.

The remainder of this paper is organized as follows: Section II reviews related work. Section III describes the data sources. Section IV explains the pipeline architecture. Section V presents results from the Banke District, Nepal case study. Section VI discusses limitations and future work. Section VII concludes.

II. RELATED WORK

Global urban mapping has evolved significantly over the past decades, progressing from pixel-based classification of Landsat imagery to large-scale machine learning products trained on extensive labeled datasets. Early approaches primarily relied on spectral thresholding techniques applied to medium-resolution satellite data, where indices such as the Normalized Difference Vegetation Index (NDVI) and Normalized Difference Built-up Index (NDBI) were used to distinguish urban from non-urban areas. While computationally efficient, these methods are highly sensitive to threshold selection and environmental conditions, often leading to inconsistent performance across different

landscapes. In particular, spectral confusion between built-up areas and bare soil, as well as seasonal variations in vegetation, limits their robustness in heterogeneous regions.

To address these limitations, supervised machine learning methods such as Random Forest classifiers have been widely adopted for land cover classification. These approaches leverage multiple spectral and spatial features and have demonstrated improved classification accuracy compared to traditional index-based methods. However, their effectiveness is strongly dependent on the quality and representativeness of the training data. Models trained on a specific geographic region often exhibit poor generalization when applied to areas with different land cover characteristics, making them less suitable for large-scale, automated pipelines spanning diverse regions.

At a global scale, several standardized land cover products have been developed to support long-term urban and environmental monitoring. The Global Human Settlement Layer (GHSL), developed by the Joint Research Centre of the European Commission, provides consistent built-up surface maps at 100 m resolution for multiple epochs between 1975 and 2020 using multi-temporal Landsat composites and machine learning classification techniques. GHSL adopts a broad definition of urban areas and offers a valuable historical baseline for long-term change analysis. However, its relatively coarse spatial resolution limits its ability to capture fine-scale urban dynamics.

Other global datasets, such as MODIS-based land cover products, provide consistent temporal coverage but at even coarser spatial resolutions (e.g., 500 m), making them unsuitable for detailed local-scale analysis. Additionally, the lack of historical data prior to the 2000s restricts their applicability for studies requiring long-term retrospective analysis.

Recent advances in deep learning have enabled the development of high-resolution, near-real-time land cover products. Dynamic World represents a significant step forward in this direction, providing 10 m resolution land cover predictions derived from Sentinel-2 imagery using a convolutional neural network trained on globally distributed reference data. This dataset offers improved spatial detail and temporal frequency, with strong performance in detecting built-up areas in well-defined urban environments. Nevertheless, deep learning approaches typically require large volumes of high-quality labeled data and may still face challenges in generalizing across diverse geographic and environmental conditions.

Several studies have explored the integration of multi-source datasets, particularly combining historical products such as GHSL with high-resolution modern datasets like Dynamic World, to enable continuous monitoring of urban expansion across different time periods. While this approach leverages the complementary strengths of each dataset, challenges remain in ensuring consistency, scalability, and automation across varying areas of interest.

Urban expansion is not a spatially random process; it follows recognizable morphological patterns governed by underlying socioeconomic, infrastructural, and physical drivers [10]. Understanding these patterns is essential for interpreting change detection outputs and informing planning interventions.

Five principal growth models are widely recognized in urban morphology literature. Concentric growth describes the outward spread of built-up area from a central urban core in roughly uniform rings, as formalized by Burgess [8]. Axial growth occurs when development extends preferentially along major transport corridors — highways, railways, and rivers producing a star-shaped urban footprint [9]. Leapfrog development is characterized by discontinuous expansion, where new built-up patches emerge beyond the existing urban fringe, separated from the core by undeveloped land; this pattern is commonly associated with speculative land development and weak land-use regulation [11]. Edge city formation refers to the emergence of commercial and industrial clusters at the urban periphery, often around road intersections or market nodes, which subsequently attract residential development [11]. Finally, infill development describes the densification of already-urbanized areas by filling vacant or underutilized land within the existing built-up envelope [10]. These patterns are not mutually exclusive; real urban expansions typically exhibit a combination of modes simultaneously in different sectors of the same city.

The primary drivers of these patterns can be grouped into four categories. First, transport and road infrastructure exerts the strongest spatial pulls on development, as accessibility reduces commuting costs and raises land values along corridors [9]. Second, land surface characteristics including terrain slope, flood risk, and soil suitability constrain where built-up expansion is physically viable, with flat, well-drained land being preferentially converted [12]. Third, economic gravity concentrates growth near markets, employment centers, and administrative nodes, reproducing spatial inequalities in development intensity [13]. Fourth, infrastructure incentives such as electrification, piped water, and government investment in peri-urban zones act as catalysts that trigger rapid land conversion in previously undeveloped areas [11].

In the context of the Terai region of Nepal, where the present study is situated, axial growth along highway corridors combined with leapfrog conversion of agricultural land has been identified as the dominant expansion mode in previous studies [12], [13]. The directional growth analysis presented in Section V is interpreted against this theoretical framework.

Overall, existing methods highlight a trade-off between spatial resolution, temporal coverage, and model generalization. Classical spectral methods lack robustness, machine learning models suffer from limited transferability, and deep learning approaches are data-intensive and computationally demanding. These limitations underscore the need for a scalable, area-independent pipeline that can integrate heterogeneous datasets while maintaining consistency across both spatial and temporal domains. The present study addresses this gap by proposing an automated workflow that combines historical and modern land cover products into a unified, reproducible framework for urban analysis.

Overall, existing methods highlight a trade-off between spatial resolution, temporal coverage, and model generalization. Classical spectral methods lack robustness, machine learning models suffer from limited transferability, and deep learning approaches are data-intensive and computationally demanding. These limitations underscore the need for a scalable, area-independent pipeline that can integrate heterogeneous datasets while maintaining consistency across both spatial and temporal domains. The present study addresses this gap by proposing an automated workflow that combines historical and modern land cover products into a unified, reproducible framework for urban analysis.

.

III. Materials and Methods

1.DATA SOURCES

A. GHSL GHS_BUILT_S (JRC, 2023 Release)

The GHS_BUILT_S product (P2023A release) provides built-up surface area in m² per 100 m pixel, derived from multi-temporal Landsat composites using a Random Forest / SVM ensemble. Data are available for epochs 1975, 1980, 1985, 1990, 2000, 2010, 2015, and 2020. The pipeline accesses this product through the Google Earth Engine image collection JRC/GHSL/P2023A/GHS_BUILT_S and exports the built surface band for the required epoch.

B. Google Dynamic World V1

Dynamic World V1 [4] provides per-pixel land cover labels at 10 m resolution from Sentinel-2 imagery with near-real-time latency. The product includes nine classes: water, trees, grass, flooded vegetation, crops, shrub and scrub, built area, bare ground, and snow and ice. The pipeline composites DW over a user-defined date range using mode aggregation to reduce ephemeral classification noise, filtered to the October–April dry season to minimize cloud contamination in South Asian study areas.

C. Landsat Surface Reflectance

Landsat 5 TM Collection 2 Tier 1 Level-2 (LANDSAT/LT05/C02/T1_L2) and Landsat 8/9 OLI Collection 2 Tier 1 Level-2 are used to generate true-colour and false-colour composites for visualization purposes only. Cloud masking uses the QA_PIXEL bitfield; scaling applies the standard USGS Collection 2 coefficients (×0.0000275 − 0.2). Landsat data are not used for classification.

2. PIPELINE ARCHITECTURE

The pipeline consists of four sequential stages orchestrated by a master runner script. All configuration is encapsulated in a single user-editable file (user_config.py); all other parameters are derived automatically. Fig. 1 illustrates the overall pipeline architecture.

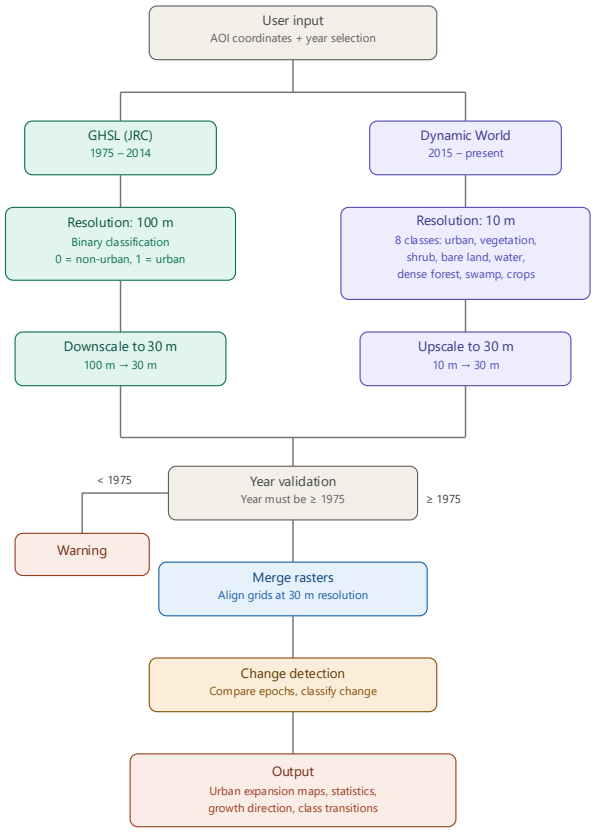

Fig. 1. *Overall pipeline architecture: four-stage workflow from GEE data acquisition through classification to change detection and output generation.*

A. Automated GEE Data Export

The pipeline authenticates with the Google Earth Engine Python API, constructs image collection queries based on the configured AOI and temporal parameters, and downloads output GeoTIFFs directly to local disk via the getDownloadURL API. The export scale is than fixed at 30 m to provide a common spatial resolution for cross-epoch comparison. All exports apply the configured AOI polygon as a clip mask.

Method selection is automatic: if YEAR_HISTORICAL < 2015, a GHSL export is initiated for the nearest available epoch; if YEAR_RECENT ≥ 2015, a Dynamic World mode composite is exported over the configured date range. Landsat composites for both epochs are always exported to support visual interpretation of the classification results.

B. Land Cover Classification

The Second stage reads the GHSL and Dynamic World rasters downloaded in Stage I and, where the AOI exceeds the single-tile download limit, merges tiles using rasterio before classification. After the initial preprocessing, it produces classified land cover rasters using one of two methods, selected automatically by epoch.

GHSL classification: The built surface band contains continuous values in m² of built-up area per pixel. A threshold of 250 m² is applied, consistent with the GHSL product definition and established practice in the urban mapping literature [3]. Pixels above the threshold are assigned Class 1 (Urban/Built-up). All remaining valid pixels are assigned Class 3 (Grass/Shrub) as a conservative non-urban background, since no spectral information is available for full seven-class mapping in the historical epoch.

Dynamic World classification: The mode-composited DW label raster contains integer class codes (0–8). These are remapped to the pipeline's seven-class scheme using the correspondence shown in Table I. NaN values (export nodata) are handled prior to integer casting to prevent overflow artefacts.

TABLE I

Dynamic World to Pipeline Class Mapping

DW code	DW Labels	Pipeline Class	Code	Colour
0	Water	Water	5	Blue
1	Trees	Trees	2	DarkGreen
2	Grass	Grass	3	LimeGreen
3	Flooded	Wetland	6	Green
4	Crops	Cropland	4	Yellow
5	Shrub	Shrub	3	LimeGreen
6	Built	Urban / Built-up	1	Red
7	Bare	Bare / Sand	7	Brown
8	Snow/ Ice	Bare / Sand	7	Brown

C. Resolution alignment and classification

For the historical epoch, the GHSL raster downloaded from GEE already contains binary values *(1 = urban, 0 = non-urban)*, having been threshold server-side at the native 100 m resolution before export. Classification therefore requires only reading the binary band and assigning class 1 (Urban/Built-up) to all pixels with a value ≥ 0.5, with all remaining valid pixels assigned class 3 (Grass/Shrub) as a non-urban background.

For the recent epoch, the Dynamic World label raster contains integer class codes (0–8). These are remapped to the pipeline's seven-class scheme: Table 1. The shrub class (DW class 5) is merged with grass (DW class 2) into a single grass/shrub category, consistent with the GHSL non-urban background class used for the historical epoch, ensuring comparability between epochs.

D. Change Detection and Output Generation

This stage loads the classified land cover rasters produced in Stage III and performs change classification, directional analysis, and statistical reporting.

Per-pixel change classification. Each pixel is assigned one of four mutually exclusive change states: stable non-urban (class 0), stable urban (class 1), new urban growth (class 2), and urban loss (class 3). New urban growth pixels are defined as pixels that were non-urban in the historical epoch and urban in the recent epoch. Urban loss pixels are the inverse. The resulting change raster is written to disk as a GeoTiffs for further GIS analysis.

Directional growth analysis. New urban growth pixels are binned into eight cardinal and intercardinal sectors (N, NE, E, SE, S, SW, W, NW) based on each pixel's angle from the AOI centroid, computed as arctan2(x − cx, cy − y). The total new urban area in km² is summed per sector and visualised as a polar bar diagram — the growth direction rose — with the dominant growth direction identified as the sector with the greatest area.

Kernel density estimation. Spatial hotspot maps are generated by applying a Gaussian filter (σ = 15 pixels, equivalent to a 450 m smoothing radius at 30 m resolution) to the binary urban masks for both epochs and to the new urban growth mask. This produces continuous density surfaces that highlight areas of concentrated urban development and are robust to classification noise at individual pixel level. Four panels are produced: historical urban density, recent urban density, growth hotspot concentration, and recent urban concentration at a broader scale (σ = 20 pixels).

Class transition statistics. For each of the seven land cover classes, the area of pixels that transitioned to urban between the two epochs is computed, quantifying the source land cover of all new urban growth. Results are presented as a horizontal bar chart and recorded in the summary statistics file.

Statistical reporting. All quantitative results are written to a structured CSV file containing: urban area for each epoch (km²), new growth (km²), urban loss (km²), net change (km²), percentage change, dominant growth direction, and classification method for each epoch. The compound annual growth rate is computed as:

$$CAGR = \left(\frac{A_2}{A_1}\right)^{\frac{1}{t}} - 1$$

where A_1 and A_2 are the historical and recent urban areas in km² respectively, and Δt is the epoch separation in years. This formulation accounts for compounding and is preferred over the simple linear rate for multi-decadal comparisons.

IV. RESULTS

The pipeline was applied to Banke, Bardiya and Rupandehi Districts in Nepal's Inner Terai, a rapidly urbanizing lowland region surrounding agriculture land and hills. The study AOI spans shows the historical urbanization and recent urbanization with land that saw the change.

TABLE II

Land Cover Area Statistics

Region	1985	2023	Growth
Banke	27 km²	102 km²	251.8%
Rupandehi	18 km²	133 km²	331.25%
Bardiya	27 km²	51 km²	74.2%

B. Spatial Distribution of Growth

Fig. 2 shows the three-panel change map for the Banke AOI. Urban expansion is concentrated along the Nepalgunj–Kohalpur corridor and along national highway corridors to the northeast. The directional rose diagram (Fig. 3) confirms northeast as the dominant growth direction, consistent with road-led expansion patterns observed in Terai urban centres.

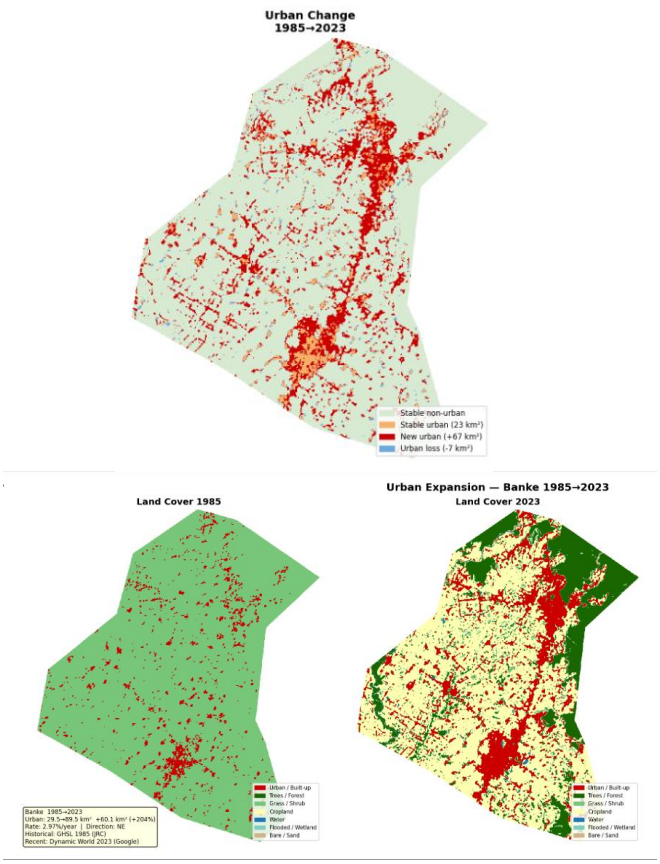

Fig. 2. *Three-panel urban expansion map: 1985 land cover, 2023 land cover, change map showing stable urban, new urban growth and urban loss*

The results shows accurate setting in context of chosen areas. The fig 2. Accurately shows the change in Kohalpur being a village back in 1985 to seeing some fast changes since 2010 after the development of a centric highway and teaching hospital. Making Nepalgunj and Kohalpur a urban concentrated cities in Banke District connect by the Surkhet road Highway.

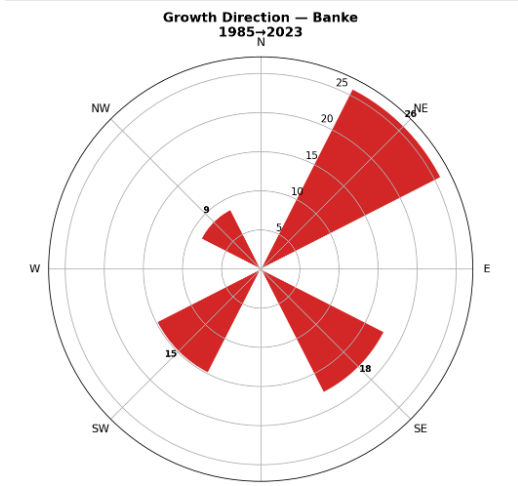

Fig. 3. *Urban growth direction rose diagram for Banke District (1985–2023). Each sector shows area of new urban growth in km². Northeast sector is dominant.*

C. Kernel Density and Growth Hotspots

Fig. 4 presents four kernel density panels: historical urban density, recent urban density, growth hotspots, and recent urban concentration. Growth hotspots are most intense at the Nepalgunj city core periphery and along the eastern highway corridor, suggesting peri-urban infill and ribbon development as the primary growth modes.

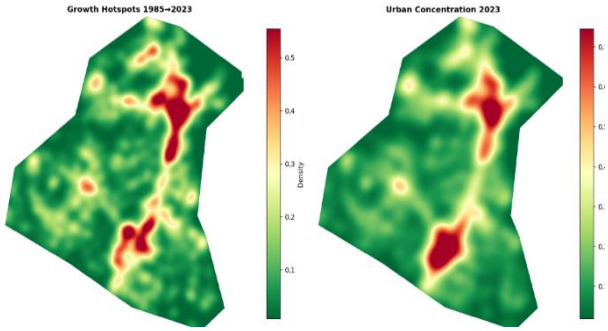

Fig. 4. *Urban kernel density heatmaps: 1985–2023 growth hotspots, 2023 urban concentration.*

D. Source Land Cover of Urban Growth

Fig. 5 shows the source land cover of new urban growth between 1985 and 2023. Around 67.6 km² of newly urbanised land is attributed to the non-urban background class. Full multi-class transition analysis identifying whether growth occurred on cropland, forest, or grassland would require both epochs to use a product with complete land cover classification or manual tunning, such as Dynamic World.

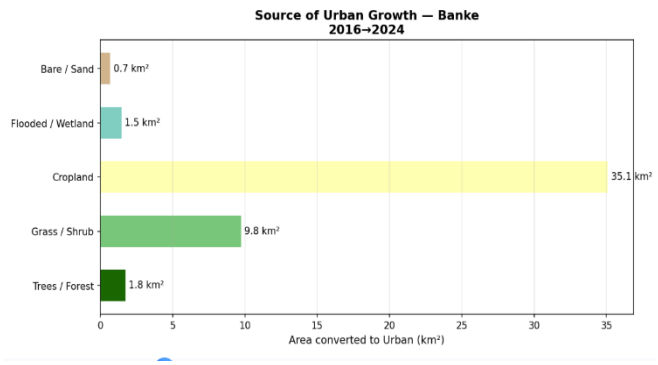

Fig. 5. *Changed pixels to urban bar graph*

V. DISCUSSION

A. Validation

The GHSL 2020 product estimates 124 km² of built-up surface for the Banke AOI. The pipeline's Dynamic World 2023 estimate of 81.5 km² is lower, reflecting the methodological difference between the two products. GHSL uses a broad definition that captures any surface with built material exceeding 250 m² per pixel, including sparse or mixed-use structures. Dynamic World applies a stricter definition focused on clearly built and impervious surfaces. Both estimates are methodologically correct within their respective frameworks, and the discrepancy is consistent with published comparisons of the two products [6].

The reported CAGR of 2.49% yr^{-1} falls within the range of published estimates for Terai urban centers (1.8–3.4% yr^{-1}) [7], providing additional confidence in the pipeline's output. Visual inspection of the change map confirms that newly classified urban areas correspond to settlement expansion visible in high-resolution imagery from Google Maps.

B. Limitations

Several limitations should be acknowledged. First, the GHSL epoch for 1985 was composited from Landsat imagery collected over 1984–1986; there is therefore a degree of temporal uncertainty in the historical map. Second, GHSL does not provide full seven-class land cover for the historical epoch; only the urban/non-urban distinction is reliable, which limits interpretation of class transition statistics for pre-2015 to-urban conversions.

While the pipeline produces results consistent with independent estimates, the use of globally trained products — GHSL and Dynamic World — as the sole classification source represents an inherent methodological ceiling. Both products are optimised for global applicability rather than regional precision, and neither incorporates region-specific spectral characteristics of the Terai landscape. For studies requiring higher classification accuracy, supervised training of a dedicated classifier using manually labelled reference points and regionally appropriate spectral band combinations, particularly the shortwave infrared and red-edge bands sensitive to built-up surface materials in South Asian contexts would yield more precise urban delineation.

C. Reproducibility and Extensibility

The pipeline is designed to be fully reproducible from a single configuration file. Users need only specify the AOI name, bounding coordinates, historical and recent. All other parameters — UTM zone, classification method, file paths, and output naming — are derived automatically. The pipeline supports any AOI globally and any historical year for which GHSL data are available (1975, 1980, 1985, 1990, 2000…) paired with any recent year from 2015 onwards.

VI. CONCLUSION

This paper has presented an automated, Four-stage pipeline for multi-temporal urban land cover change detection that leverages the complementary strengths of the JRC Global Human Settlement Layer and Google Dynamic World. By selecting the most appropriate globally peer-reviewed product for each temporal epoch, the pipeline achieves consistent, training-data-free classification that is applicable to any AOI worldwide. Applied to Banke District, Nepal, the pipeline produced urban growth estimates consistent with independent data sources and published regional statistics.

Future work will extend the pipeline's analytical capabilities beyond retrospective change detection toward predictive urban modelling. By integrating population growth data with the directional expansion patterns identified in this study, it may be possible to estimate the trajectory and rate of urbanisation in specific geographic directions. Such analysis could serve as a foundation for urban growth forecasting models, with practical applications in real estate valuation, infrastructure investment planning, and land-use policy — particularly in rapidly urbanising regions such as the Terai where spatial expansion patterns are strongly corridor-driven.

REFERENCES

[1] K. C. Seto, B. Güneralp, and L. R. Hutyra, “Global forecasts of urban expansion to 2030 and direct impacts on biodiversity and carbon pools,” Proc. Natl. Acad. Sci., vol. 109, no. 40, pp. 16083–16088, 2012.

[2] Q. Weng, “Remote sensing of impervious surfaces in urban areas: Requirements, methods, and trends,” Remote Sens. Environ., vol. 117, pp. 34–49, 2012.

[3] M. Pesaresi et al., “Atlas of the Human Planet 2023 — JRC Technical Report,” Joint Research Centre, European Commission, Luxembourg, 2023.

[4] C. F. Brown et al., “Dynamic World, near real-time global 10 m land use land cover mapping,” Sci. Data, vol. 9, no. 1, p. 251, 2022. doi: 10.1038/s41597-022-01307-4.

[5] P. Gong et al., “Annual maps of global artificial impervious area (GAIA) between 1985 and 2018,” Remote Sens. Environ., vol. 236, p. 111510, 2020.

[6] D. Taati and M. Hashemi, “Comparison of GHSL and Dynamic World for urban extent mapping: A multi-city evaluation,” Int. J. Remote Sens., vol. 44, no. 12, pp. 3812–3831, 2023.

[7] Central Bureau of Statistics Nepal, “National Population and Housing Census 2021: Urban Statistics,” Government of Nepal, Kathmandu, 2022.

[8] E. W. Burgess, "The growth of the city: An introduction to a research project," in The City, R. E. Park, E. W. Burgess, and R. D. McKenzie, Eds. Chicago, IL, USA: Univ. of Chicago Press, 1925, pp. 47–62.

[9] N. Baum-Snow, "Did highways cause suburbanization?" Quarterly Journal of Economics, vol. 122, no. 2, pp. 775–805, May 2007.

[10] B. Bhatta, Analysis of Urban Growth and Sprawl from Remote Sensing Data. Berlin, Germany: Springer, 2010.

[11] S. Angel, J. Parent, D. L. Civco, and A. M. Blei, Planet of Cities. Cambridge, MA, USA: Lincoln Institute of Land Policy, 2012.

[12] R. B. Thapa and Y. Murayama, "Urban growth mapping of Kathmandu Metropolitan Region using remote sensing and GIS," Sensors, vol. 9, no. 6, pp. 4441–4458, Jun. 2009.

[13] T. G. McGee, "The emergence of desakota regions in Asia: Expanding a hypothesis," in The Extended Metropolis: Settlement Transition in Asia, N. Ginsburg, B. Koppel, and T. G. McGee, Eds. Honolulu, HI, USA: Univ. of Hawaii Press, 1991, pp. 3–25.

[14] S. Uddin, M. R. Hossain, and S. K. Halim, “Urban growth dynamics of Kathmandu Valley using remote sensing and GIS,” Environ. Monit. Assess., vol. 188, no. 7, pp. 1–15, 2016.

[15] R. Thapa and Y. Murayama, “Examining spatiotemporal urbanization patterns in Kathmandu Valley using remote sensing and spatial metrics,” Remote Sens., vol. 2, no. 7, pp. 1901–1918, 2010.

[16] B. Rimal et al., “Urban expansion and agricultural land loss in Kathmandu Valley, Nepal,” Land Use Policy, vol. 81, pp. 185–195, 2019.

[17] A. Paudel and T. Yuan, “Spatiotemporal analysis of land use/land cover change in Nepal using Landsat data,” Sustainability, vol. 4, no. 11, pp. 1–18, 2012.

[18] J. Liu et al., “A spatially explicit approach to modeling urban expansion in China using cellular automata and neural networks,” Landscape Urban Plan., vol. 85, no. 3–4, pp. 247–259, 2008.